Molecular Catalysis 550 (2023) 113570

Contents lists available at ScienceDirect

Molecular Catalysis

journal homepage: www.journals.elsevier.com/molecular-catalysis

Efficient whole-cell biosynthesis of (*S*)-2-chloro-1-(3,4-difluorophenyl)-ethanol from 2-chloro-1-(3,4-difluorophenyl) ethanone in a sustainable reaction system

Linghui Kong [1], Bo Fan [1], Yu-Cai He [*]

School of Pharmacy & School of Biological and Food Engineering, Changzhou University, Changzhou 213164, China

ARTICLE INFO

Keywords:
(*S*)-2-Chloro-1-(3,4-difluorophenyl)-ethanol
2-Chloro-1-(3,4-difluorophenyl) ethanone
Asymmetric bioreduction
Reductase
Deep eutectic solvent

ABSTRACT

(*S*)-2-Chloro-1-(3,4-difluorophenyl)-ethanol [(*S*)-CFPL] is a key chiral intermediate in the manufacture of Ticagrelor. In this study, deep eutectic solvent (DES) was used as the co-solvent, and several reaction parameters were investigated and characterized. It was observed that supplementary of DES could promote the asymmetric reduction of 2-chloro-1-(3,4-difluorophenyl) ethanone (CFPO) to (*S*)-CFPL through the biological reduction by recombinant *E. coli* containing a NADH-dependent reductase CmCR. The high pKa value of carboxylic acid in DES and low viscosity of DES were beneficial to increase cell membrane permeability and enhancement of biocatalytic efficiency. By supplementary of DES ChCl:LA (7.5 vol%) into the bioreaction system, the efficient bioreduction reaction of 100 mM CFPO was achieved via the biocatalysis with whole-cell at 30 °C and pH 7.0, affording (*S*)-CFPL in 95.9 % yield and optical purity > 99 %. The process was environmentally friendly with high biocatalytic productivity and had potential practicability in the mass production of (*S*)-CFPL.

1. Introduction

Optically active chiral alcohols have received a lot of attention due to their relatively stable molecular structure and their potential as key building-block in the manufacture of materials, pesticides and various compounds used in pharmaceuticals [1]. For example, (*S*)-1-(2, 6-dichloro-3-fluorophenyl) ethanol is an important intermediate for Crizotinib [2]. Other chiral alcohols are also important intermediates of Duloxetine [3], Aprepitant [4], and Sitagliptin [5], as shown in Fig. 1. Chiral compounds are now synthesized by biocatalysts [6–8]. Compared with chemical catalysis, biocatalysis has the characteristics of high efficiency [9,10], specificity [11,12], gentleness [13] and adjustable activity [14]. Ticagrelor is a platelet aggregation inhibitor primarily used in acute coronary syndromes [15], and (*S*)-2-chloro-1-(3,4-difluorophonyl)-ethanol [(*S*)-CFPL] is a key chiral intermediate for preparing Ticagrelor. The chemical synthesis of (*S*)-CFPL involves the reduction of CFPO in organic solvents by chemocatalysts such as borane-methyl sulfide complex and chiral compounds [16]. Compared with biocatalysis, traditional chemical reduction methods always involve a variety of expensive chiral catalysts and cumbersome post-treatment processes [17]. Therefore, the synthesis method of (*S*)-CFPL needs to be further improved. Currently, two methods to synthesize chiral alcohols are commonly utilized via biocatalytic approach: dynamic kinetic resolution and asymmetric bioreduction [18]. Han et al. identified an alcohol dehydrogenase from *Rhodococcus* R6 that catalyzed the asymmetric transformation of 2-hydroxyacetophenone into (*R*)-(-)-1-phenyl-1,2-ethanediol [19]. Xia et al. obtained three optical compounds by developing a lipase from *Candida Antarctica* that catalyzed the doubly enantioselective ring-opening decomposition between racemic alcohol and lactone [20]. Tozlu et al. used *Lactobacillus paracasei* BD101 as a biocatalyst for asymmetric reduction of (4-chlorophenyl)(phenyl) methane [21]. Chemical catalyst can cause major ecological problems in terms of high energy consumption and toxic waste formation. Biocatalysis has outstanding advantages over chemocatalysis, including mild reaction conditions, high stereoselectivity and high catalytic efficiency [22]. Biocatalyst is one of the attractive technology options to overcome the limitations of traditional catalysts in this field.

During the process of biocatalysis, reaction medium is very important to improve the biocatalytic efficiency [23]. More and more attention has recently been paid to deep eutectic-solvents (DESs) in biocatalysis [24–26]. DES is commonly composed of

* Corresponding author.
E-mail addresses: heyucai2001@163.com, yucaihe@cczu.edu.cn (Y.-C. He).
[1] Author contribution: They contributed equally to this work.

https://doi.org/10.1016/j.mcat.2023.113570
Received 30 July 2023; Received in revised form 15 September 2023; Accepted 16 September 2023
Available online 7 October 2023
2468-8231/© 2023 Elsevier B.V. All rights reserved.

hydrogen-bond-donor (HBD) and hydrogen-bond-acceptor (HBA). Due to its good biocompatibility and high stability, it is considered as the most promising green solvent to replace ionic liquid and organic solvent [27]. DES synthesized from choline chloride is one of the many types of DESs, which could change the composition by adjusting different HBDs [28,29]. DESs have been widely utilized in the chemocatalysis and biocatalysis [30]. They have important physical properties such as viscosity, conductivity, and thermal stability [31–33]. In addition, the typical characteristics of DES in the process of biocatalytic reaction are as follows: (a) higher solubility for substrates with poor water solubility, which make the substrates more easily recognized and utilized by biocatalysts [34]; (b) improve the activity and stability of biocatalysts [35]; (c) reduce the occurrence of side reactions, which make the reaction more selective enantiomers [36]. DES may alter the viscosity of bioreaction medium, and the change in osmolality caused by the viscosity will influence the permeability of cell membrane. The enhanced permeability of bacterial cell membranes (by mutations in its structure) might lead to a significant improvement in whole-cell biocatalytic efficiency [37].

Distinct from the isolated enzymes, one of the main advantages of the whole-cell biocatalysts is that the whole cells provide an excellent natural environment for the enzymes, allow co-factors to regenerate, and avoid enzyme inactivation and denaturation that might occur under some harsh performance conditions [38–41]. The whole cell catalyst is also the cheapest catalyst formulation. DES is a new type of green, solvent with low volatility, high stability, good biocompatibility and other characteristics [33]. In this study, whole cells of recombinant *E. coli* containing a NADH-dependent reductase CmCR [42] were used for bioreduction of CFPO into (*S*)-CFPL by using the mixture of DES and H_2O as reaction system. The relationship of pKa value of carboxylic acid in DES, viscosity of DES and biocatalytic activity was explored. The effects of DES as a co-solvent on cell viability and cell membrane permeability were also studied, and then the effects of DES on biotransformation were also examined. In addition, biological reaction parameters (such as reaction temperature, pH, co-substrate loading, cofactor loading, metal ion loading and biocatalyst loading) were investigated. Finally, a sustainable production strategy for biocatalytic synthesis of ticagrelor chiral intermediates (*S*)-CFPL was developed in a benign reaction system.

2. Materials and methods

2.1. Chemicals and materials

CFPO was obtained from Aladdin Co., Ltd. (Shanghai, China). Choline chloride (ChCl), tartaric acid, malonic acid, citric acid, *D,L*-malic acid, lactic acid, *D*-glucose, KH_2PO_4, K_2HPO_4, $MgCl_2$, and other reagents were from Shanghai Lingfeng Chemical Reagent Co., Ltd. (China).

2.2. Synthesis of DESs

Equal molar of ChCl as HBA and organic acid (tartaric acid, malonic acid, citric acid, malic acid, and lactic acid) as HBD were supplemented to a 250-mL flask by stirring magnetically in an oil bath at 80 °C and heated for 2 h. ChCl:Tartaric acid (ChCl:TA), ChCl:Malonic acid (ChCl:MOA), ChCl:Citric acid (ChCl:CA), ChCl: Malic acid (ChCl:MA) and ChCl:Lactic acid (ChCl:LA), were used as cosolvent.

2.3. Determination of Kamlet-Taft parameters of DESs

The measurement of Kamlet-Taft (K-T) parameters were determined, with *N,N*-dimethyl-4-nitroaniline as indices of π*, Nile red dye and 4-nitroaniline as indices of α and β, respectively. Dyes were prepared from anhydrous methanol solution, in which the concentrations of *N,N*-dimethyl-4-nitrobenzene, Nile red and 4-nitroaniline were all 1 mM. The dye solution (2 mL) was first transferred to a 5 mL centrifuge tube, then vacuum-dried at 40 °C for 5 h to remove the methanol. 3 mL DES was added to the test tube for dissolving the dye under the ultrasound (300 W) at 25 °C, and then transferred to a quartz colorimetric dish. Ultraviolet spectrophotometer (L6S, Shanghai INESA Scientific Instruments Co., Ltd) was applied to detect the absorption wavelength of the mixture in the wavelength range of 300–700 nm at room temperature. The solvatochromic parameters were calculated using the following equations:

$$V = \frac{1}{\lambda_{max} \times 10^{-4}} \tag{1}$$

$$\alpha = \frac{19.9657 - 1.0241\pi^* - V_{\text{Nile red}}}{1.6078} \tag{2}$$

$$\beta = \frac{1.035V_{N,N-\text{dimethyl}-4-\text{nitrobenzene}} + 2.64 - V_{4-nitroaniline}}{2.80} \tag{3}$$

$$\pi^* = 0.314 \times \left(27.52 - V_{N,\ N-\text{dimethyl}-4-\text{nitrobenzene}}\right) \tag{4}$$

Where λ_{max} represents the maximum absorption wavelength.

2.4. Preparation of racemic alcohols

A 250 mL round-bottomed flask was filled with 20 mM CFPO, 80 mL anhydrous ethanol and 30 mM sodium borohydride. This preparation was carried out in an ice water bath with magnetic agitation. Thin layer

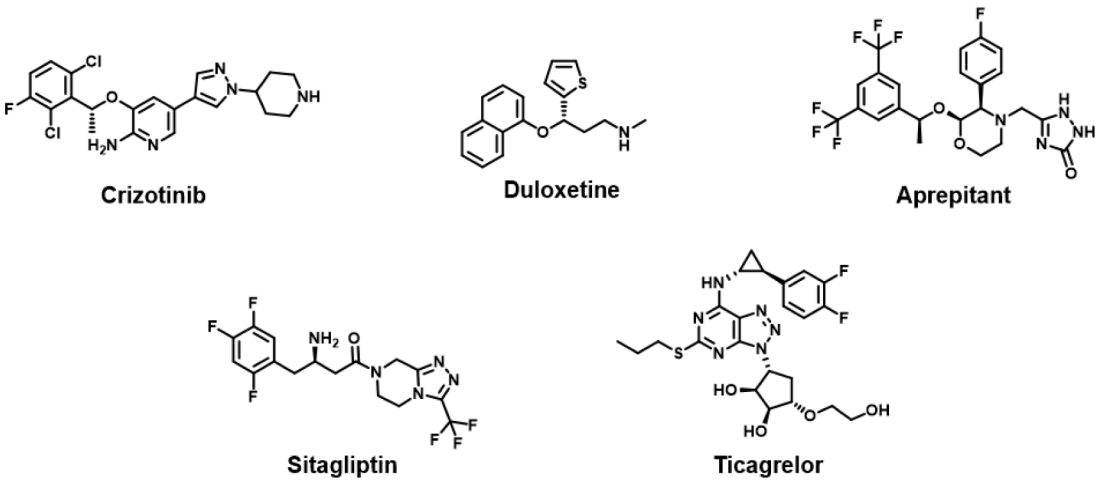


Fig. 1. The structure of Crizotinib, Duloxetine, Aprepitant, Sitagliptin and Ticagrelor.

chromatography (TLC) was used to detect the reaction progress. After the transformation of substrate was complete, 5 M hydrochloric acid was slowly added drop by drop to pH 4.0 to stop the reaction. The salt was removed from the system by pumping and filtration, and the ethanol was removed by rotary evaporation. The prepared solution was extracted with ethyl acetate, and a small amount of anhydrous magnesium sulfate was supplemented into the organic phase for drying for 1 h, then the ethyl acetate was removed by filtration and vacuum distillation. The prepared racemic alcohol was stored at 4 °C, and the purity of the product was analyzed by gas chromatography (GC).

2.5. Molecular docking

Three kinds of *E. coli* containing carbonyl reductase CmCR [47], SsCR [39] and *Kp*ADH [40] were preserved in the laboratory, namely *E. coli* CCZU-K14, *E. coli* CCZU-SF21 and *E. coli Kp*ADH. The most suitable enzyme was selected by molecular docking. CmCR was a model structure based on Genbank: BAE46987.1, built through a Swiss model server. SsCR was a NADPH-dependent carbonyl reductase (PDB: 1ZZE) from *Sporospora salmoni*. *Kp*ADH was a polysporous alcohol dehydrogenase (PDB: 5Z2X) of *Kluyveromycetes*. The substrate CFPO was connected to the model structure of CmCR, SsCR and *Kp*ADH via AutoDock Vina. For molecular docking, ligands and residues were set to be flexible and rigid, respectively. The size and center Settings of the grid box were as follows: CmCR: center x, y, $z = 1.159, 0.37, -1.127$, size x, y, $z = 40, 40, 40$; SsCR: center x, y, $z = 3.546, 15.0, 19.997$, size x, y, $z = 40, 40, 40$; *Kp*ADH: center x, y, $z = -46.344, 123.84, -1.362$, size x, y, $z = 40, 40, 40$. The docking run was conducted, and the substrate for docking with the direction of low energy production was selected for analysis.

2.6. Measurement of reductase activity

The reductase activity was determined using 2.0 mM CFPO as substrate by monitoring the decrease in the absorbance of NADH at 340 nm. The reaction mixture used to determine reductase enzyme activity contained 2.95 mL buffer (potassium phosphate buffer, pH 7.0), 0.10 mM NADH, 2.0 mM CFPO and 50 μL crude enzyme (ultrasonic broken) in a total volume of 3.0 mL. One unit (U) of reductase activity was defined as the amount of enzyme catalyzing 1.0 μmol NADH per minute under these conditions.

2.7. Determination of membrane integrity

The integrity of the cell membrane was assessed by EU-2200 UV spectrophotometer (Shanghai Onlab Instruments Co., Ltd) with absorbance values at 260 nm and 280 nm. The whole cells were completely suspended in the buffer solution and a certain amount of DES was added under 30 °C and 180 rpm for 8 h. The cloudy cell fluid was extracted, the cells were rotated and centrifuged, and the OD_{260} and OD_{280} values were determined. Three parallel controls were performed for each experiment.

The membrane integrity of the cells was measured by flow cytometry (FCM) and fluorescence microscopy (FL). 100 mg/mL of cells were incubated in the reaction media under 30 °C, pH 7.0 and 180 rpm for 8 h. Then it was diluted to 150 CFU/mL with PBS of pH 7.0, stained with 1 mL propidium iodide (PI) (50 μg/mL), and stored at 4 °C in the dark for 30 min. The integrity of the cell membrane was detected by flow cytometry (BD Accuri™ C6 Plus) with a wavelength of 488 nm and an emission wavelength of 630 nm. The cell morphology was observed by fluorescence microscope (Carl Zeiss, ZEN-3–0-blue-Hotfix-4) with magnification of 30 times.

2.8. Bioreduction of CFPO

Recombinant *E. coli strains* were cultured at 37 °C in LB medium by supplementary of kanamycin (50 μg/mL). When cell OD_{600} (optical density at 600 nm) reached 0.60, isopropyl-β-*D*-thiogalactoside (IPTG) was supplemented into the culture media with an induction content of 0.50 mM. After incubating at 25 °C for 12 h, cells were washed and centrifuged in 50 mM phosphate buffer with pH 7.0 (8000 × g, 8 min), and finally collected and used for bioreduction.

In this study, several bioreaction parameters was tested on the bioreduction of CFPO. In the reaction system containing potassium phosphate buffer (100 mM, pH 6.0–8.0), CFPO was biologically catalyzed with glucose (0–3.0 mol/mol CFPO), NAD^+ (0–3.0 μM/mM CFPO), DESs (HBA: ChCl; HBD: TA, MOA, MA, CA and LA) dosage (0–50, vol%), 100.0 mM CFPO, whole cells (0.1 g/mL), metal ions (Zn^{2+}, Sn^{4+}, Ba^{2+}, Al^{3+}, Ca^{2+}, Mn^{2+}, Cu^{2+}, Mg^{2+}, Ni^{2+}, and Fe^{2+}) (1.0 mM). After incubation in a shaker (180 rpm) for 0–8 h under the temperature of 30 °C, CFPO and (*S*)-CFPL then detected by HPLC.

The solubility of CFPO was measured at pH 7.0 and 30 °C in DES-water. In the mixture of ChCl:LA (7.5 vol%) and water (92.5 vol%), the solubility of CFPO was 44.2 g/L. In the mixture of ChCl:MA (7.5 vol %) and water (92.5 vol%), the solubility of CFPO was 51.8 g/L. In the mixture of ChCl:CA (7.5 vol%) and water (92.5 vol%), the solubility of CFPO was 38.7 g/L. In the mixture of ChCl:TA (7.5 vol%) and water (92.5 vol%), the solubility of CFPO was 46.3 g/L. In the mixture of ChCl: MOA (7.5 vol%) and water (92.5 vol%), the solubility of CFPO was 30.6 g/L. To test the influence of substrate CFPO concentrations on the stability of reductase. 50–600 mM commercial CFPO, a certain amount of glucose and NAD^+, and whole cells (0.1 g/mL, wet weight) were mixed together in the mixture of ChCl:LA (7.5 vol%) and water (92.5 vol%) (pH 7.0) under the temperature of 30 °C for 0–24 h. The production of (*S*)-CFPL was calculated by extracting samples periodically, extracting the (*S*)-CFPL produced by ethyl acetate and the unreacted CFPO, and analyzing them using HPLC.

2.9. Analytical methods

The viscosity of DESs were measured by a rotary viscometer (NDJ-1C, Shanghai FangRui Instrument Co., Ltd) at 25 °C. FT-IR (Fourier Transform Infrared Spectroscopy) was detected on Thermo Scientific Nicolet iS20 (USA) over 4000–500 cm^{-1}. ^1H NMR (H Nuclear Magnetic Resonance) was carried out by Bruker Avance III HD (400 MHz). DES (5 mg) was dissolved in 250 μL DMSO for ^1H NMR detection. The *ee* value of (*S*)-CFPL was measured with GC (model 8890; Agilent Technologies Co., Ltd) containing a CP-Chirasil-Dex-CB column (25 m × 0.25 mm, 0.25 μm) with N_2 as the carrier gas. The substrate CFPO and product (*S*)-CFPL were measured with HPLC (Agilent 1260) with an Athena C18-WP column (4.6 × 250 mm, 5 μm) at 260 nm using acetonitrile/water (80:20, v/v) as the eluent at a flow rate of 0.8 mL per minute. The (*S*)-CFPL yield and *ee* value were defined as below:

$$(S)-CFPL(\%) = \frac{\text{CFPL formed (mM)}}{\text{Inital CFPO (mM)}} \times 100\% \quad (5)$$

$$ee\ (\%) = \frac{(S)-\text{CFPL} - (R)-\text{CFPL}}{(S)-\text{CFPL} + (R)-\text{CFPL}} \times 100\% \quad (6)$$

3. Results and discussion

3.1. Molecular docking screening reductase

Molecular docking could be applied to assess the interaction between ligand and receptor, and evaluate the optimal binding mode in the region of receptor active site according to the principle of spatial structure complementarity and energy minimization. The ligand of CFPO was predicted by molecular docking simulation using Auto dock and PyMOL software. The interactions between CFPO ligands and important residues of CmCR were displayed in Fig. 2a and b. The binding site had two hydrogen-bonds and residues interacting with TYR-165 and SER-127. As illustrated in Fig. 2b, the bonding distance between CFPO and the

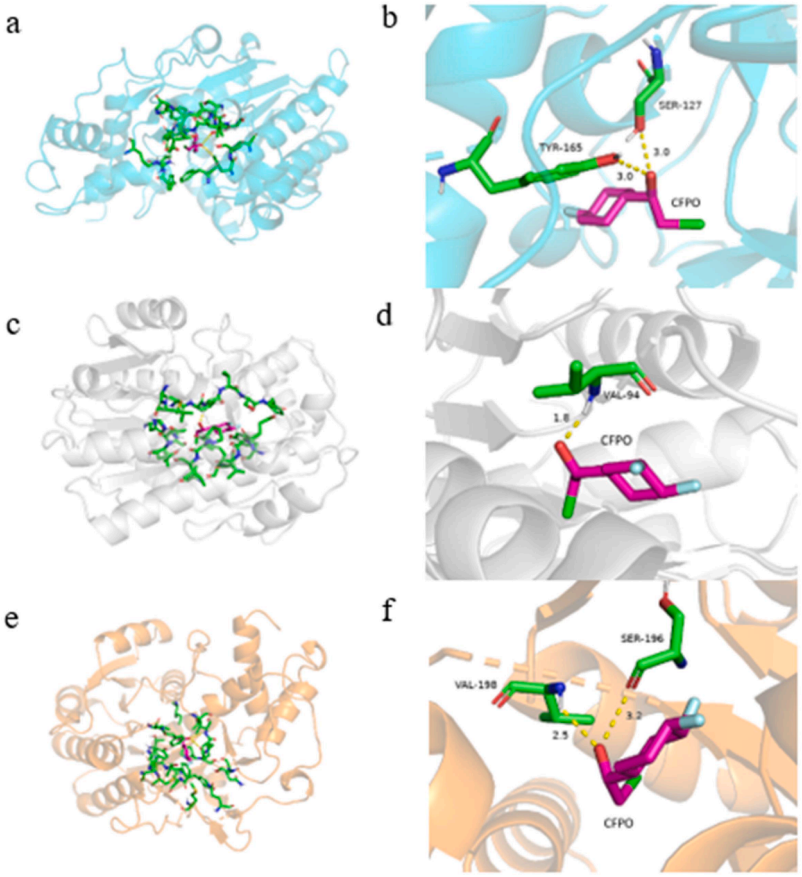


Fig. 2. Molecular docking images [Molecular docking of reductase CmCR with CFPO molecules] (a); Two amino acids hydrogen-bonded to CFPO molecules (b); Molecular docking images [Molecular docking of reductase SsCR with CFPO molecules] (c); One amino acid hydrogen-bonded to CFPO molecules (d); Molecular docking images [Molecular docking of reductase *Kp*ADH with CFPO molecules] (e); Two amino acids hydrogen-bonded to CFPO molecules (f).

interacting amino acid (AA) was calculated (3.0 Å for TYR-165 and 3.0 Å for SER-127). The interaction of CFPO ligands with important AA residues of CmCR was presented in Fig. 2c and d. There was a hydrogen-bond and VAL-94 interaction residue at the binding site. As depicted in Fig. 2d, the bonding distance between CFPO and the interacting AA was 1.9 Å for VAL-94. The interaction of CFPO ligands with important AA residues of *Kp*ADH was revealed in Fig. 2e and f. The binding site had two hydrogen-bonds and residues interacting with VAL-198 and SER-196. As indicated in Fig. 2f, the bonding distance between CFPO and the interacting AA was 2.5 Å for VAL-198, 3.2 Å for SER-196.

It was seen from Table 1 and Fig. 2 that CFPO at the binding site of CmCR had a binding energy of −6.7 kcal/mol, which was the lowest energy and the most suitable for biological reduction of CFPO. Consequently, *E. coli* harboring CmCR was selected as CFPO-reducing biocatalyst in the subsequent experiments.

3.2. Biological reduction of CFPO with whole cells by supplementary of DES

Different from chemical reduction, biological reduction has attracted extensive attention because it can be carried out under the ambient conditions, avoiding the use of precious metal catalysts [43]. According to reports, most of the biological reduction of CFPO uses organic solvent-water as the reaction system [44]. In order to improve the sustainable transformation of CFPO through the biological reduction, it was necessary to establish a green reaction medium for the transformation of CFPO to (*S*)-CFPL.

Table 1
CmCR, SsCR, and *Kp*ADH molecule docking CFPO molecules, the number of hydrogen bonds and the minimum binding energy.

Reductase	Bonding hydrogen bonds	Binding energy (kcal/mol)
CmCR	2	−6.7
SsCR	1	−5.7
*Kp*ADH	2	−6.4

In previous studies, DESs had been used as solvents and chemocatalysts for the green synthesis of small molecules which showed excellent cell biocompatibility [45]. DESs also exhibit good biocatalytic activity in asymmetric reduction [46]. Compared with the isolated enzymes, whole-cell biocatalysts could be prepared more easily and are inexpensive [47]. In this study, whole cells biologically reduced CFPO was supplemented to the DES-H_2O system for synthesis of (*S*)-CFPL. Using ChCl was as HBA, five carboxylic acids [Tartaric acid (TA, pKa = 3.04), Malonic acid (MOA, pKa = 2.82), Citric acid (CA, pKa = 3.13), Malic acid (MA, pKa = 3.46), and Lactic acid (LA, pKa = 3.86)] were used as HBDs for preparing DESs (ChCl:TA, ChCl:MOA, ChCl:CA, ChCl:MA, and ChCl:LA). These DESs were individually mixed with water to form DES-H_2O (1:9, v/v) system for enhancing the bioreduction of CFPO into (*S*)-CFPL (Fig. 3a). It was observed that the (*S*)-CFPL yield increased significantly in the existence of LA-mediated DES, and a higher (*S*)-CFPL yield (90.8 %) was achieved, which was 1.4-fold higher than that in the aqueous system. The relative activity was 1.9 times that of single-phase water system. ChCl:LA was measured with ^{1}H NMR and FT-IR (Figs. S1 & S2 in Supporting Information): ChCl:LA [^{1}H NMR (400 MHz, DMSO-d_6) δ 4.04 (1H, LA, CH_2-C), 3.81 (2H, ChCl, N-CH_2), 3.45 (2H, ChCl, C-CH_2), 3.08 (9H, ChCl, N-CH_3), 1.20 (3H, LA, CH_2-OH)]. The peak at 3158 cm^{-1} was due to -OH bond vibration in ChCl:LA. The peak at 1718 cm^{-1} was ascribed to C=O skeleton vibrations in ChCl:LA. The

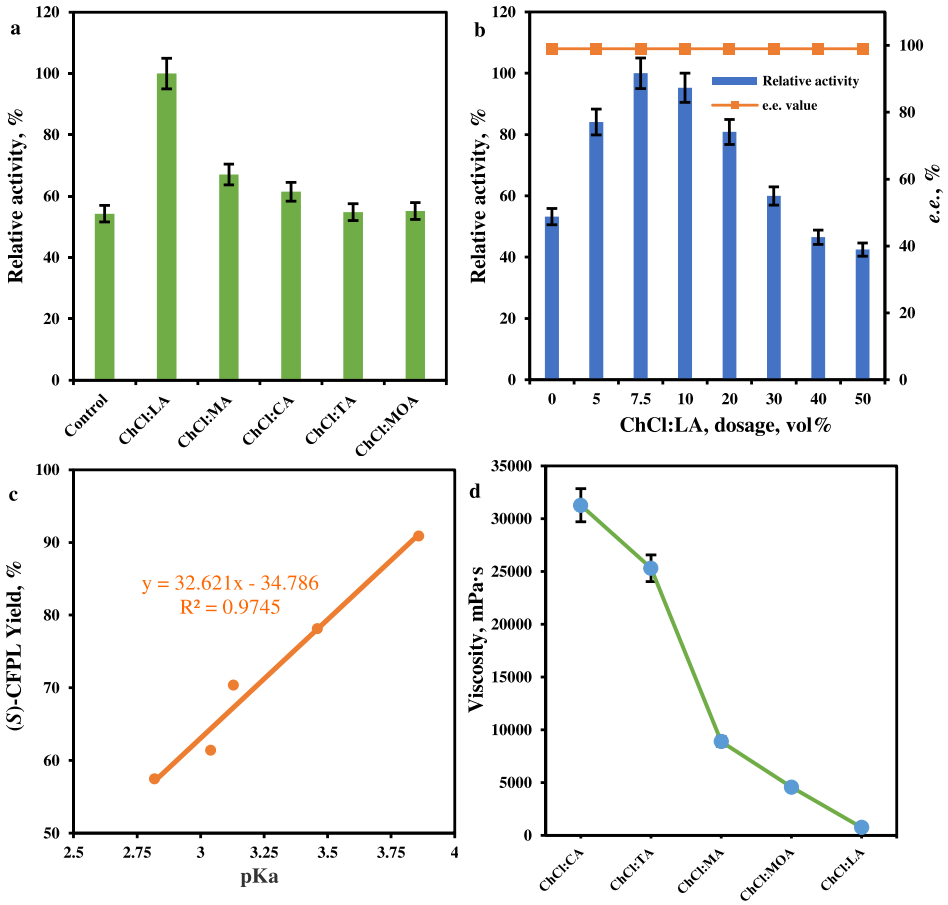


Fig. 3. Effects of different DES on bioreduction activity by using the mixture of DES (10 vol%) and H_2O (90 vol%) as reaction system [100 mM CFPO, 0.1 g/mL whole cells, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 30 °C, 180 rpm, 30 min] (a); Effect of ChCl:LA dose on bioreduction activity by using the mixture of ChCl:LA and H_2O as reaction system [ChCl:LA (0–50 vol%), 100 mM CFPO, 0.1 g/mL whole cells, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 30 °C, 180 rpm, 30 min] (b); Linear relationship between the pKa value of carboxylic acid in DES with the yield of (*S*)-CFPL by using the mixture of DES (10 vol%) and H_2O (90 vol%) as reaction system [100 mM CFPO, 0.1 g/mL whole cells, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 30 °C, 180 rpm, 8 h] (c); Viscosity values of different DES (d).

peak at 958 cm^{-1} was attributed to the slight vibration of the C-N group in ChCl:LA. In addition, water could bridge HBA and HBD, and an appropriate amount of water could improve the stability of the DES-H_2O system [48]. The addition of 10.1 wt% water to DES consisting of ChCl and glycerol strengthened the direct hydrogen-bond between ChCl and glycerol [49]. Hence, the different volumetric ratio of ChCl:LA-to-water might greatly affect the biological reduction of CFPO to (*S*)-CFPL. In DES ChCl:LA-H_2O, the volumetric ratio range of 0–50 v% ChCl:LA was used (Fig. 3b). When the volumetric ratio of ChCl:LA increased from 0 to 7.5 v %, the relative activity gradually increased from 65.2 % to 93.3 %. As the ratio of ChCl:LA increased from 7.5 to 10 v%, the relative activity did not increase apparently. The relative activity was weakened obviously when ChCl:LA was more than 10 v%. Excessive water loading in ChCl:LA-H_2O might cause changes in osmotic pressure of the solution, which not only facilitated contact between reductase and substrates in the whole cells, but also reduced potential product or substrate inhibition, thus enhancing biocatalytic efficiency [50]. Apparently, the optimum dose of ChCl:LA was 7.5 % by volume, and ChCl:LA-water (7.5:92.5, v/v) was used in subsequent investigation. The *ee* value of (*S*)-CFPL was detected by GC, reaching over 99 % (Fig. S3 in Supporting Information).

The ionization of hydrogen ions in DES might affect the synthesis of small organic molecules, suggesting that the acidity of carboxylic acid groups would affect the production of (*S*)-CFPL [51]. In the biological of reduction of CFPO, a linear relationship [$Y_{[(S)\text{-CFPL}],\%}$] = 32.621 × $X_{(pKa\ of\ carboxylic\ acid)}$ - 34.786, R^2=0.9745] was observed between the pKa of carboxylic acid and the yield of (*S*)-CFPL (Fig. 3c). The high pKa of carboxylic acid in DES would promote the synthesis of (*S*)-CFPL. When pKa increased from 2.82 to 3.86, the yield of (*S*)-CFPL increased linearly, which might be related to the increase of proton utilization rate. High pKa value of HBD, the weaker its acidity, facilitated the whole-cell bioreduction of CFPO. From the structural characteristics of α-carboxylic acid of LA, it could be seen that stronger H-bonding formed after, protonation and ion-dipole interaction was established between DES and water, increasing the freedom degree of bioreduction activity and further improving the bioreduction efficiency of the bioreaction system [52].

In the process of biocatalytic reaction, different viscosity of DESs might affect the efficiency of biocatalytic reaction [53]. Therefore, the viscosity of five DES was tested (Fig. 3d). The results indicated that the viscosity of DES (ChCl:LA) was relatively low. High viscosity of DES will cause a great damage degree of cell membrane, leading to a small leakage of material inside the cell membrane, thereby limiting mass transfer [54]. In DES composed of acetamide, urea and sorbitol, the lower the viscosity of the reaction medium, the higher the activity of bromelain [55].

3.3. *Investigation of cells membrane permeability by supplementary of DES*

The cell membrane is the first target of the solvent [56]. When the

cell permeability is enhanced, its components (protein and nucleic acid) can be easily released. The change in the efficiency of whole-cell biocatalysis suggested a certain interaction between cells and DESs. Compared with the aqueous system, DESs showed higher absorbance at 260 nm and 280 nm (Fig. 4a). The results showed that DESs could increase the permeability of cells membrane within 8 h. In addition, the ability of DES to extend the cell membrane was enhanced according to the sequence of ChCl:MOA < ChCl:TA < ChCl:CA < ChCl:MA < ChCl:LA. This phenomenon showed that DESs could increase the permeability of cells, which was not only beneficial to the contact between enzymes and substrates in cells, but also helped alleviate product inhibition, thus improving the biocatalytic efficiency. Zhu et al. had demonstrated that the reaction for whole-cells synthesis of (*S*)−1-[3,5-bis(trifluoromethyl) phenyl] ethanol introduced by DES (choline chloride:trehalose, 1:1 molar ratio) [57], DES had the ability to promote cell membrane permeability and facilitate the contact between substrates to reductase in cells.

Damaged cells and dead cells could be distinguished by flow cytometry, and PI was used as a fluorescent dye. As shown in Fig. 4b, compared with the blank, the proportion of damaged cells in DES-water system increased obviously. It was found that the proportion of damaged cells in the mixture of ChCl:LA and water reached up to 17.6 %. Generally speaking, DES enhanced the permeability of cell membrane, which was beneficial for the full contact between substrate and enzyme, but it also led to cell death due to contact with toxic substances. From the fluorescence microscope (FL), it could be seen more directly (Fig. S4 in Supporting Information) that the addition of DES had damaged the cell membrane to a certain extent, and it could be seen that whole cells added with DES would have more division, resulting in increased staining of damaged cells. This may be because the addition of DES increases the viscosity of the reaction system, leading to the increase of cell membrane damage.

Cao et al. investigated the permeability of hydrophobic DES to cell membranes and applied hydrophobic DESs to release intracellular enzymes from recombinant *Escherichia coli* [58]. As displayed in Fig. 5a, DES composed of ChCl and lactic acid (molar ratio of 1: 1) was used as catalyst and solvent with acidic and ionic properties. Because DES changed the osmotic pressure and viscosity of the reaction system, there was a little leakage of proteins and nucleic acids on the cell membrane, which improved the biocatalytic efficiency. The asymmetric reduction of CFPO to (*S*)-CFPL by whole cells was a process of consuming NADH, and the regeneration of NADH by glucose dehydrogenase formed a coupling cycle with the reduction reaction, so that the reaction could be carried out continuously. The whole reaction process was presented in Fig. 5b.

3.4. Polarity analysis of DESs

The Kamlet-Taft solvatochromic method could be used as a quantitative tool to estimate the chemical properties of two component carboxylic acids DESs. Kamlet-Taft parameters π*, α and β represent the hydrogen-bond dipolarity/polarizability, acidity and basicity of the solvent, respectively [59]. DESs with a high α value could easily provide hydrogen-bonds to liberate H^+ and have a high acidity. DESs with a high β value had higher hydrogen-bond acceptance capacity. The α and β values of ChCl:LA were 0.51 and 0.53, respectively, indicating that the acidity and hydrogen-bond acceptability of the solvent were low (Table 2). However, the initial reaction rate in ChCl:LA-H_2O system was the highest, reaching 1.52 mM/min. This might be due to the fact that the system had fewer active protons, reduced the acidic site, and made the pH value of the solvent more neutral, which was more suitable for whole-cells to participate in biological reduction.

3.5. Enhancement of biological reduction from CFPO to (S)-CFPL by using the mixture of ChCl:LA and H_2O as reaction system

It is well known that bioreaction temperature, medium pH, metal ion additive, co-substrate, and cofactor could influence the bioreduction greatly during the biological transformation [60–62]. Thus, it was necessary to optimize biological reduction factors (such as glucose content, cofactor NAD^+ loading, metal ion, reaction temperature, reaction pH, etc.) in order to enhance the bioreduction efficiency. The performance temperature and medium pH have a crucial influence on the bioreaction [63]. As displayed in Fig. 6a, the highest CFPO-reducing activity was observed at 30 °C. Beyond this performance temperature, the bioreduction activity declined gradually. When the performance temperature increased, the three-dimensional structure of reductase changed, thus lowering the activity of reductase. When the temperature was too high, the structure of the enzyme molecules would change irreversibly, making the enzyme inactive [64]. The highest reduction activity was observed at pH 7.0 (Fig. 6b). When pH deviated from the optimum value, the activity of reductase would be inhibited or

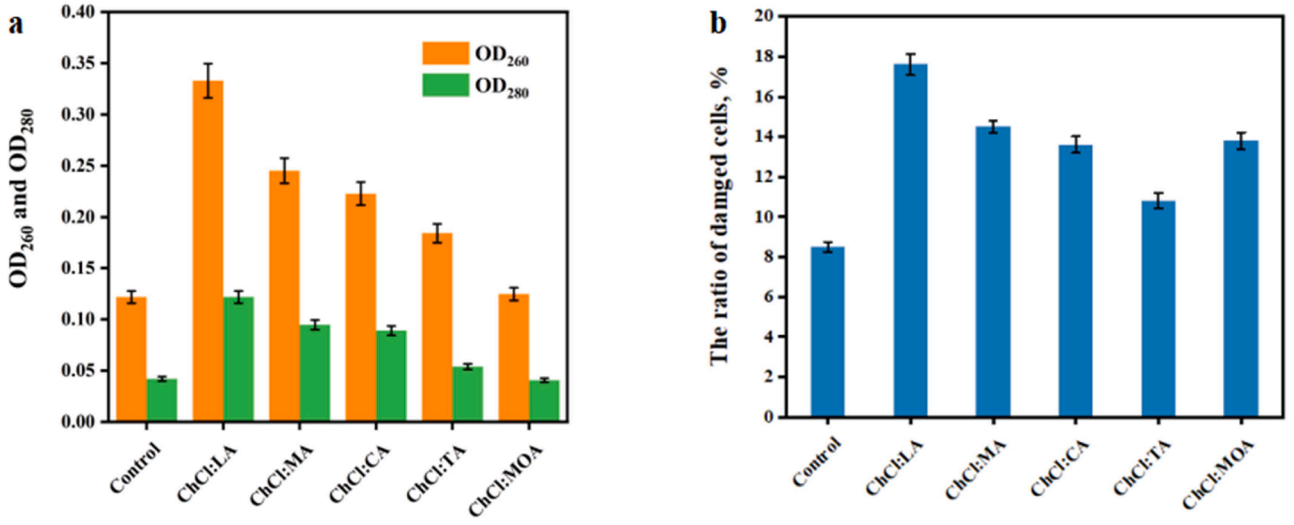


Fig. 4. Effects of on the membrane integrity of whole cells were analyzed over ultraviolet spectrophotometer [The membrane permeability of cells after incubation in kalium phosphate buffer (KPB) (pH 7.0) or the mixture of DES (7.5 vol%) and H_2O (92.5 vol%), 30 °C, 180 rpm] (a); Effects of KPB system or the mixture containing DES (7.5 vol%) and H_2O (92.5 vol%) on the ratio of damaged cells [Treatment condition: 30 °C, 8 h, pH 7.0, 180 rpm; DES species: (ChCl:LA, ChCl:MA, ChCl:CA, ChCl:TA, and ChCl:MOA)] (b).

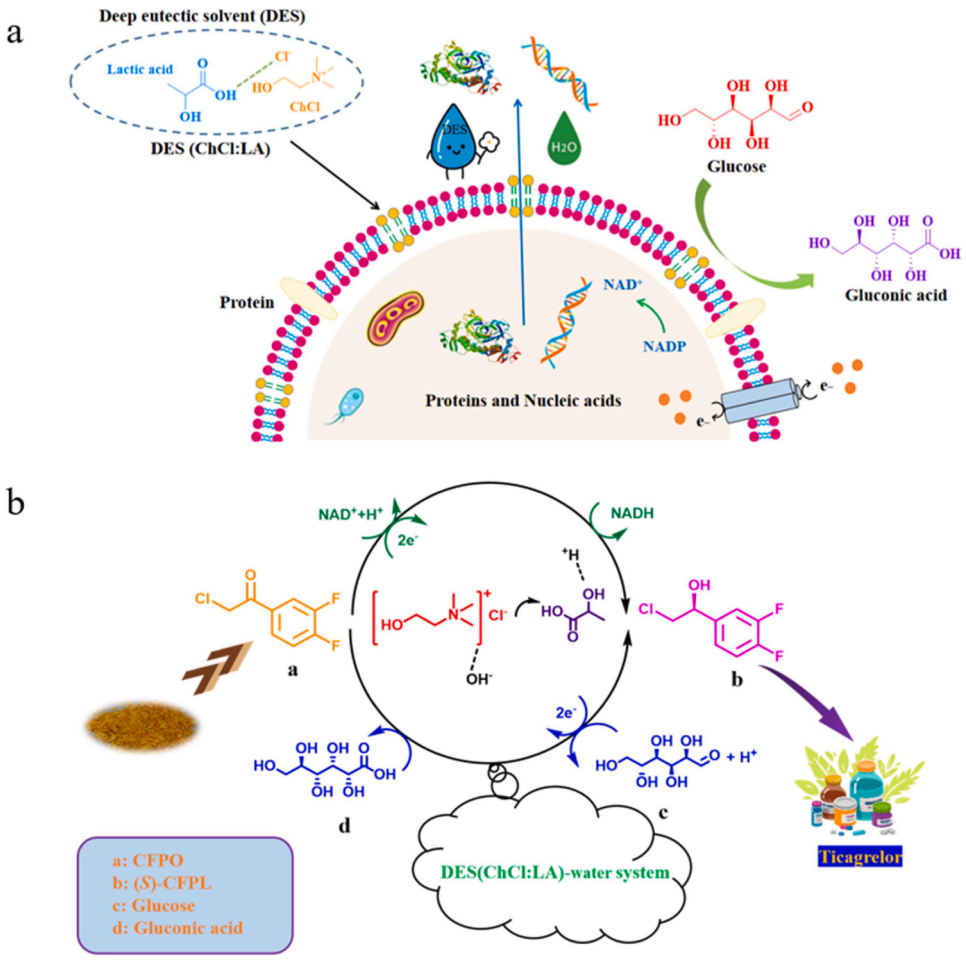


Fig. 5. Mechanism diagram of cell membrane permeability in DES-water systems (a); Biological reduction of CFPO to (*S*)-CFPL in DES-water (b).

Table 2
Kamlet-Taft parameters of two-component deep eutectic solvent.

Experiment No.	HBA/HBD Molar Ratio	DES name	Kamlet-Taft parameters			Initial reaction rate[a] (mM/min)
			π*	α	β	
1	1:1	ChCl:TA	1.30	0.83	1.31	0.85
2	1:1	ChCl:MOA	1.34	0.65	1.23	0.83
3	1:1	ChCl:CA	1.08	0.69	0.20	0.96
4	1:1	ChCl:MA	1.20	0.70	0.26	1.04
5	1:1	ChCl:LA	1.13	0.51	0.53	1.52

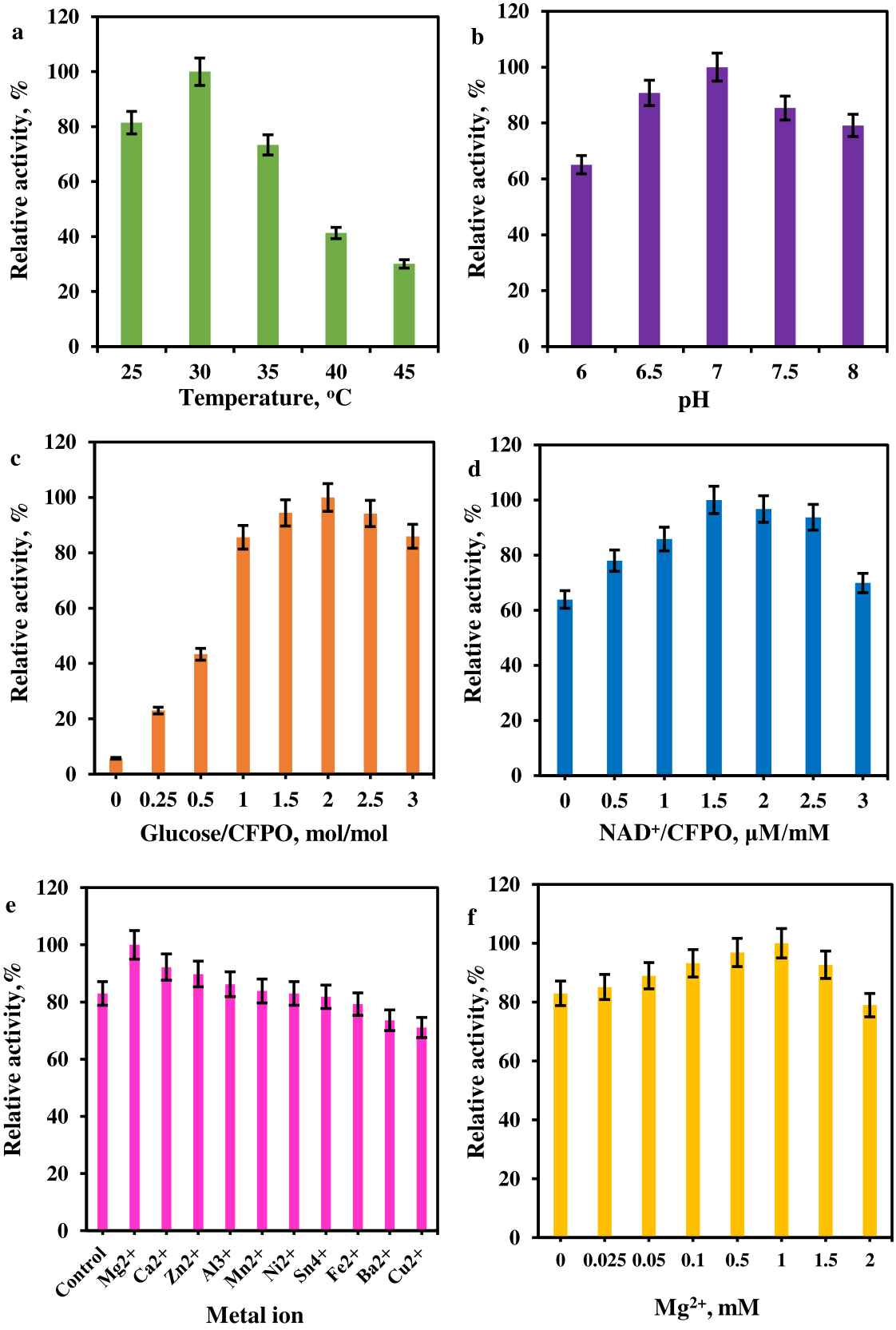
[a] Initial reaction rate (mM/min) [Reaction condition: DES-H_2O (7.5:92.5, v/v), 100 mM CFPO, 0.1 g/mL cells, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 30 °C, 180 rpm, 0.5 h].

decreased. This was because the change of pH value affects the affinity between the reductase and the substrate, which changes the reaction rate [65].

Glucose has been utilized as co-substrate to regenerate intracellular cofactor [66,67]. As displayed in Fig. 6c, by raising glucose load from 0 to 2.0 mol glucose per mol CFPO, the reduction activity increased greatly. Over the loading of 2.0 mol glucose per mol CFPO, the catalytic activity of CFPO was weakened gradually. The addition of excessive glucose would make the bioreaction medium more viscous, thereby causing decreased bioreduction activity [68]. Hence, 2.0 mol glucose per mol CFPO was used for improving the CFPO-reducing activity. In a bioreduction reaction system, cofactor NAD^+ is a common prerequisite for enhancing the biological reduction efficiency [69,70]. The effect of NAD^+ dose on CFPO-reducing bioreaction was displayed in Fig. 6d. Upon raising NAD^+ content from 0 to 1.5 μm NAD^+/mmol CFPO, the relative activity increased gradually. As the content of NAD^+ reached 1.5 μm NAD^+/mmol CFPO, the biocatalytic activity arrived at the maximum. After NAD^+ exceeded 1.5 μm NAD^+/mmol CFPO, the activity declined. Thus, the optimal NAD^+ load for the biological reduction of CFPO was 1.5 μm NAD^+ per mmol CFPO.

In the biocatalytic process, the addition of metal ions might stabilize the structure of intermediate and avoid the occurrence of unwanted side-reactions, thus enhancing productivity [71]. Accordingly, different chloride salts (1.0 mM) were supplemented to ChCl:LA-H_2O system at 30 °C and pH 7.0. As revealed in Fig. 6e, Ba^{2+} and Cu^{2+} had strong inhibition on the reductase activity. Mg^{2+}, Ca^{2+}, and Zn^{2+} enhanced the bioreduction activity. In particular, Mg^{2+} (0.1 mM) gave the highest activity. Effects of Mg^{2+} doses (0–2.0 mM) were examined on the reductase activity at pH 7.0 under the temperature of 30 °C (Fig. 6f). It was observed that Mg^{2+} (0.025–2.0 mM) did not inhibit the biological reduction, suggesting that Mg^{2+} might promote the whole-cell bioconversion of CFPO. The highest biocatalytic activity was detected in the existence of 1.0 mM Mg^{2+}. Probably, Mg^{2+} might combine with amino acid residues of reductase or other small molecules to form active centers, thus promoting the biocatalytic reaction. In addition, Mg^{2+} might influence the stability capacity of reductase [72].

In addition, the effects of different substrate concentrations (50–600



(caption on next page)

Fig. 6. Effect of temperature on CFPO-reducing reaction by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system [0.1 g/mL whole cells, 100 mM CFPO, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 180 rpm] (a); Effect of pH on CFPO-reducing reaction by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system [0.1 g/mL whole cells, 100 mM CFPO, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), 30 °C, 180 rpm] (b); Effects of glucose load on CFPO-reducing reaction by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system [0.1 g/mL whole cells, 100 mM CFPO, NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 30 °C, 180 rpm] (c); Effect of NAD^+ load on CFPO-reducing reaction by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system [0.1 g/mL whole cells, 100 mM CFPO, glucose (2 mol glucose/mol CFPO), pH 7.0, 30 °C, 180 rpm] (d); Effects of metal chloride salt types (1.0 mM) on CFPO-reducing reaction by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system 0.1 g/mL whole cells, 100 mM CFPO, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 30 °C, 180 rpm] (e); Effect of various concentration of Mg^{2+} on CFPO-reducing reaction by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system [0.1 g/mL whole cells, 100 mM CFPO, glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), pH 7.0, 30 °C, 180 rpm] (f).

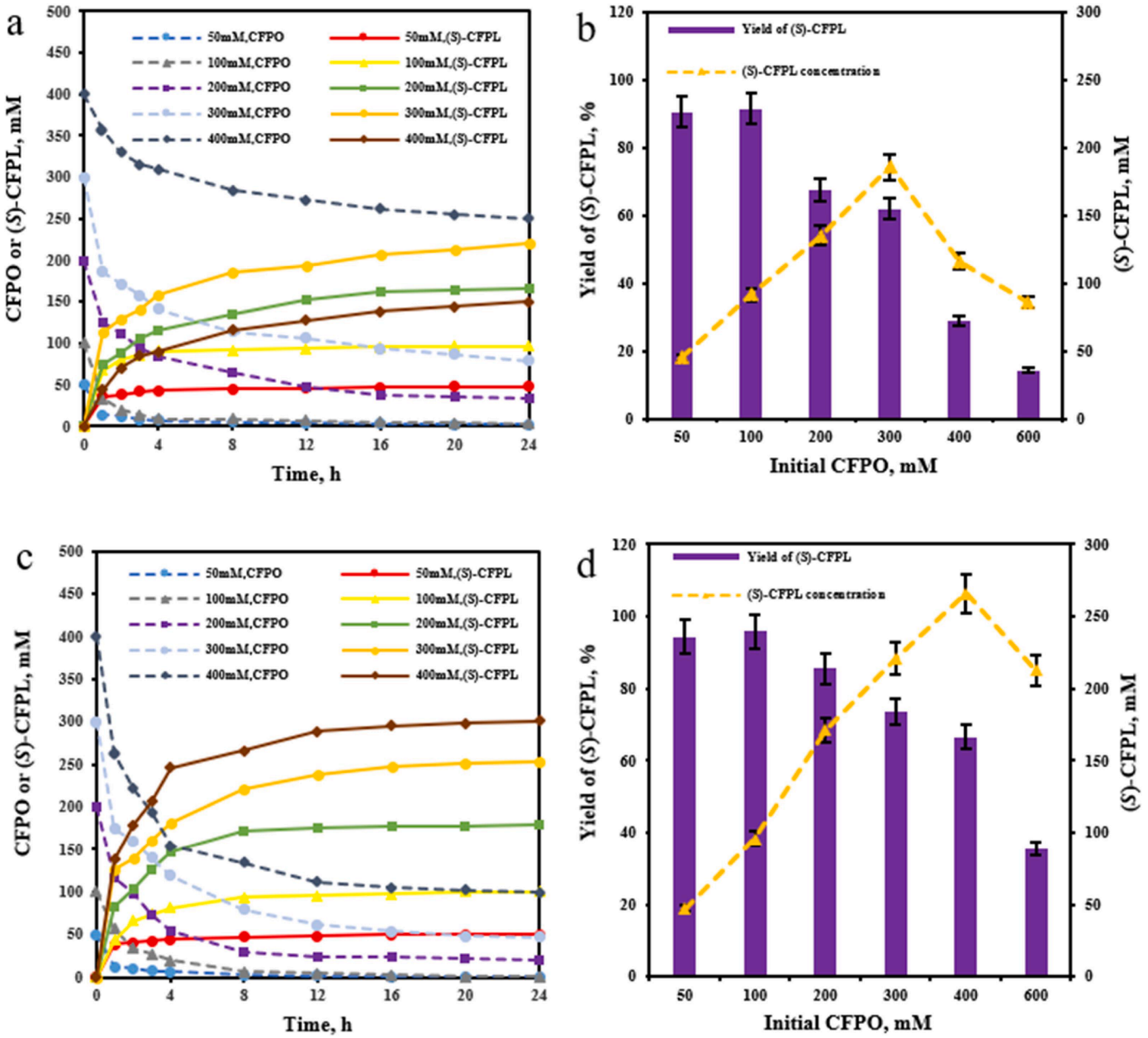


Fig. 7. Effect of initial CFPO content (50–400 mM) on the formation of (*S*)-CFPL in a water system [cells (0.10 g/mL), glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), Mg^{2+} (1.0 mM), pH 7.0, 30 °C, 180 rpm] (a); The biological reduction of CFPO (50–600 mM) in a water system [cells (0.10 g/mL), glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), Mg^{2+} (1.0 mM), pH 7.0, 30 °C, 180 rpm, 8 h] (b); Effect of initial CFPO content (50–400 mM) on the formation of (*S*)-CFPL by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system [cells (0.10 g/mL), glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), Mg^{2+} (1.0 mM), pH 7.0, 30 °C, 180 rpm] (C); The biological reduction of CFPO (50–600 mM) by using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system [cells (0.10 g/mL), glucose (2 mol glucose/mol CFPO), NAD^+ (1.5 μmol NAD^+/mmol CFPO), Mg^{2+} (1.0 mM), pH 7.0, 30 °C, 180 rpm, 8 h] (d).

mM) on bioreduction efficiency in the aqueous system were studied in Fig. 7a and b. In aqueous solution, the yield of (*S*)-CFPL decreased with the increase of substrate load. When the substrate concentration was lower than 100 mM, CFPO could be completely consumed within 24 h. Over 100 mM, the generation of (*S*)-CFPL was significantly weakened. When the CFPO was 400 mM, the (*S*)-CFPL yield dropped to below 40 % within 24 h. This might be due to the toxic effect of excessive substrate on whole cells biocatalysts, reducing their activity. By increasing the 3,5-bis(trifluoromethyl) acetophenone (3,5-BTAP) concentration from 60 mM to 120 mM, the (*R*)-(3,5-bis(trifluoromethyl)phenyl) ethanol [(*R*)-BTPE] yield was declined from 80 % to 45 % using *T. asperellum* cell as biocatalyst [23]. When the concentration of CFPO was greater than 300 mM, the concentration of (*S*)-CFPL began to decline after 8 h of biological reduction reaction.

As one new type of solvent, DES might effectively improve the whole-cell biocatalysis. By using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system, the effects of different substrate concentrations (50–600 mM) on the whole-cell bioreduction efficiency were investigated in Fig. 7c and d. 50–100 mM CFPO could be completely converted to (*S*)-CFPL within 2–8 h at 30 °C. When the CFPO was 400 mM, the (*S*)-CFPL yield reached 75 % within 24 h. When the concentration of CFPO was greater than 400 mM, the concentration of (*S*)-CFPL began to decline after 8 h of biological reduction reaction, effectively improving the substrate concentration tolerance of whole cells. Asymmetric reduction of CFPO was conducted using mutant reductase (KmCR-W_3) in DMSO-H_2O system, obtaining the yield of (*S*)-CFPL (*ee* > 99 %) within 6 h in a yield of 95.5 % [73]. The green catalytic synthesis in DES system was displayed in Fig. 8. The product (*S*)-CFPL was analyzed by ^{1}H NMR and gas chromatography-mass spectrometry (Figs. S5& S6 in Supporting Information): (*S*)-CFPL [^{1}H NMR (400 MHz, Chloroform-*d*) δ 7.28 – 7.23 (m, 1H), 7.23 – 7.15 (m, 1H), 7.10 (ddd, J = 8.2, 4.3, 1.9 Hz, 1H), 4.88 (dd, J = 8.6, 3.4 Hz, 1H), 3.72 (dd, J = 11.3, 3.4 Hz, 1H), 3.59 (dd, J = 11.3, 8.5 Hz, 1H), 2.80 (s, 1H)].

(*S*)-CFPL is a key chiral building block of Ticagrelor for the treatment of acute coronary syndrome [74]. Compared with traditional chemical synthesis, biological reduction process has the advantages of environmental friendliness, energy saving and high selectivity [75–77]. By using the mixture of ChCl:LA (7.5 vol%) and H_2O (92.5 vol%) as reaction system (pH 7.0), 100 mM CFPO could be transformed to (*S*)-CFPL (95.9 % yield, > 99 % *ee*) at 30 °C for 8 h in the existence of glucose (200 mM), NAD^+ (150 μmol), and Mg^{2+} (1.0 mM). In the biocatalytic process, the presence of DES (ChCl:LA) might promote the leakage of nucleic acid and protein in cells. In ChCl:LA-H_2O system, an efficient bioreduction of CFPO into (*S*)-CFPL was demonstrated successfully in this work. In the future, in-depth exploration will be conducted on how does the docking affect about reductase and DES components. Some binding pockets may appear in the simulation of DES cosolvent molecular docking, causing changes in the local conformation of the protein, resulting in the opening of the instantaneous pockets, thereby reducing the binding free energy. Furthermore, the cost-effective biosynthesis of (*S*)-CFPL from the high titer of CFPO with robust reductase would be developed in an eco-friendlier and more sustainable bioreduction system.

4. Conclusion

Biocatalysis is currently accepted as a one of key methodologies for manufacturing chiral drug intermediates. In this work, molecular docking revealed that CFPO at the binding site of CmCR had a binding energy of −6.7 kcal/mol, which favored the bioreduction of CFPO. The efficient asymmetric transformation of CFPO to prepare (*S*)-CFPL was successfully carried out in DES-water system through the whole cells bioreduction with *E. coli* cell harboring CmCR. It was found that the high pKa value of carboxylic acid (as HBD) in ChCl-based DES and low viscosity of DES might be beneficial to promote the cell membrane permeability of whole-cells and enhance the bioreduction efficiency. In the ChCl:LA-H_2O system, 100 mM CFPO was converted to (*S*)-CFPL with a yield of 95.9 %. This strategy provided a sustainable and efficient biocatalytic method for the effective preparation of (*S*)-CFPL.

CRediT authorship contribution statement

Linghui Kong: Methodology, Data curation, Software, Writing – original draft. **Bo Fan:** Methodology, Investigation, Data curation. **Yu-Cai He:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial

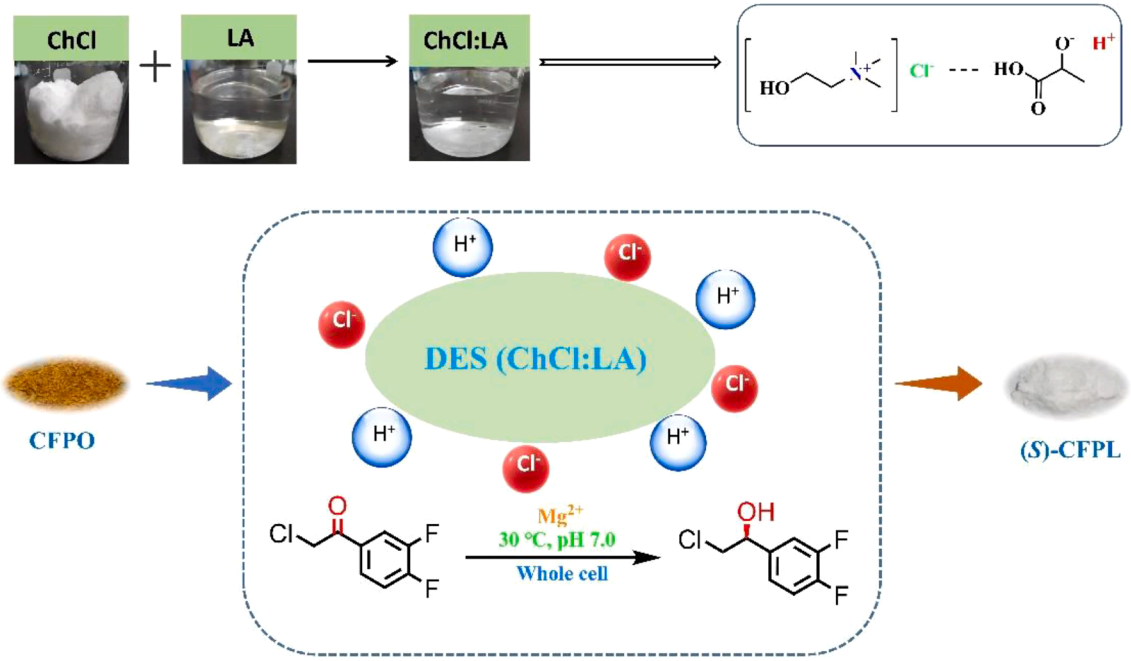


Fig. 8. Schematic diagram of green catalytic synthesis of DES system.

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This study was funded by the National Natural Science Foundation of China (no. 21978072 & 22208031) and the Open Funding Project of the State Key Laboratory of Bioreactor Engineering.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mcat.2023.113570.

References

[1] Z. Sun, J. Zhang, D. Yin, G. Xu, Y. Ni, Novel alcohol dehydrogenase CgADH from Candida glabrata for stereocomplementary reduction of bulky–bulky ketones featuring self-sufficient NADPH regeneration, ACS Sustain. Chem. Eng. 10 (2022) 13722–13732.
[2] C. Zong, X. Zhang, F. Yang, Y. Zhou, N. Chen, Z. Yang, G. Ding, F. Yu, Y. Tang, Biotransformation of a crizotinib intermediate using a mutant alcohol dehydrogenase of Lactobacillus kefir coupled with glucose dehydrogenase, Prep. Biochem. Biotechnol. 49 (2019) 578–583.
[3] Z. Ou, H. Zhao, L. Tang, W. Zhang, G. Yang, Asymmetric synthesis of duloxetine intermediate (*S*)-(-)-3-N-methylamino-1-(2-thienyl)-1-propanol using immobilized *Saccharomyces cerevisiae* in liquid-core sodium alginate/chitosan/sodium alginate microcapsules, Bioprocess Biosyst. Eng. 37 (2014) 2243–2250.
[4] J. Tang, T. Wei, G. Ni, X. Guo, Y. Wu, F. Zhang, S. Chen, Development of an efficient and cost-effective enzymatic process for production of (R)-[3,5-bis (trifluoromethyl)phenyl] ethanol using carbonyl reductase derived from Leifsonia sp. S749, Appl. Biochem. Biotechnol. 188 (2019) 87–100.
[5] Y. Wei, S. Xia, C. He, W. Xiong, H. Xu, Highly enantioselective production of a chiral intermediate of sitagliptin by a novel isolate of *Pseudomonas pseudoalcaligenes*, Biotechnol. Lett. 38 (2016) 841–846.
[6] E. Şahin, E. Dertli, Highly Enantioselective production of chiral secondary alcohols with *Candida zeylanoides* as a new whole cell biocatalyst, Chem. Biodivers. 14 (2017), e1700121.
[7] D. Yılmaz, E. Şahin, E. Dertli, Highly enantioselective production of chiral secondary alcohols using *Lactobacillus paracasei* BD101 as a new whole cell biocatalyst and evaluation of their antimicrobial effects, Chem. Biodivers. 14 (2017), e1700269.
[8] Y. Baydaş, E. Dertli, E. Şahin, Green synthesis of chiral aromatic alcohols with Lactobacillus kefiri P2 as a novel biocatalyst, Synth. Commun. 50 (2020) 1035–1045.
[9] E. Şahin, E. Dertli, Biocatalyzed Enantiomerically Pure Production of (S)-Phenyl (thiophen-2-yl)methanol, J. Heterocycl. Chem. 56 (2019) 2884–2888.
[10] E. Kalay, E. Şahin, Biocatalytic asymmetric synthesis of (R)-1-tetralol using Lactobacillus paracasei BD101, Chirality 33 (2021) 447–453.
[11] E. Şahin, Green synthesis of enantiopure (S)-1-(benzofuran-2-yl)ethanol by whole-cell biocatalyst, Chirality 31 (2019) 892–897.
[12] E. Şahin, Production of (*R*)-1-(1,3-benzodioxol-5-yl)ethanol in high enantiomeric purity by *Lactobacillus paracasei* BD101, Chirality 30 (2018) 189–194.
[13] E. Şahin, H. Serencam, E. Dertli, Whole cell application of Lactobacillus paracasei BD101 to produce enantiomerically pure (S)-cyclohexyl(phenyl)methanol, Chirality 31 (2019) 211–218.
[14] E. Kalay, E. Şahin, Regioselective asymmetric bioreduction of trans-4-phenylbut-3-en-2-one by whole-cell of Weissella cibaria N9 biocatalyst, Chirality 33 (2021) 535–542.
[15] S.J. Lee, J.Y. Cho, B.K. Kim, K.H. Yun, Y. Suh, Y.H. Cho, Y.H. Kim, A.Y. Her, S. Cho, D.W. Jeon, S.Y. Yoo, D.K. Cho, B.K. Hong, H.M. Kwon, S.J. Hong, C.M. Ahn, D. H. Shin, C.M. Nam, J.S. Kim, Y.G. Ko, D. Choi, M.K. Hong, Y. Jang, Ticagrelor monotherapy versus Ticagrelor with Aspirin in patients with ST-segment elevation myocardial infarction, JACC Cardiovasc. Interv. 14 (2021) 431–440.
[16] H. Zhang, J. Liu, L. Zhang, L. Kong, H. Yao, H. Sun, Synthesis and biological evaluation of ticagrelor derivatives as novel antiplatelet agents, Bioorg. Med. Chem. Lett. 22 (2012) 3598–3602.
[17] R.A. Sheldon, The E factor 25 years on: the rise of green chemistry and sustainability, Green Chem. 19 (2017) 18–43.
[18] Y. Huang, N. Liu, X. Wu, Y. Chen, Dehydrogenases/reductases for the synthesis of chiral pharmaceutical intermediates, Curr. Org. Chem. 14 (2010) 1447–1460.
[19] M.N. Han, X.M. Wang, C.H. Pei, C. Zhang, Z. Xu, H.L. Zhang, W. Li, Green and scalable synthesis of chiral aromatic alcohols through an efficient biocatalytic system, Microb. Biotechnol. 14 (2021) 444–452.
[20] B. Xia, Y. Li, G. Cheng, X. Lin, Q. Wu, Lipase-catalyzed doubly enantioselective ring-opening resolution between alcohols and lactones: Synthesis of chiral hydroxyl esters with two stereogenic centers, ChemCatChem 6 (2014) 3448–3454.
[21] N. Tozlu, A.S. Bülbül, A. Özdemir, E. Şahin, Efficient bio-catalytic production of enentiopure (*S*)-(4-chlorophenyl)(phenyl)methanol as a drug precursor by using a novel rotatable composite design-based optimization strategy, Mol. Catal. 547 (2023) 113404.
[22] E. Şahin, Efficient bioreduction of 1-(furan-2-yl)ethanone into enantiomerically pure drug precursor by Lactobacillus paracasei BD101, Mol. Catal. 539 (2023), 113037.
[23] J. Li, P. Wang, Y.S. He, Z.R. Zhu, J. Huang, Toward designing a novel oligopeptide-based deep eutectic solvent: Applied in biocatalytic reduction, ACS Sustain. Chem. Eng. 7 (2019) 1318–1326.
[24] A. Paiva, R. Craveiro, I. Aroso, M. Martins, R.L. Reis, A.R.C. Duarte, Natural deep eutectic solvents – solvents for the 21st Century, ACS Sustain. Chem. Eng. 2 (2014) 1063–1071.
[25] T.X. Yang, L.Q. Zhao, J. Wang, G.L. Song, H.M. Liu, H. Cheng, Z. Yang, Improving whole-cell biocatalysis by addition of deep eutectic solvents and natural deep eutectic solvents, ACS Sustain. Chem. Eng. 5 (2017) 5713–5722.
[26] M. Panić, M. Cvjetko Bubalo, I. Radojčić Redovniković, Designing a biocatalytic process involving deep eutectic solvents, J. Chem. Techno. Biotechnol. 96 (2020) 14–30.
[27] F.M. Perna, P. Vitale, V. Capriati, Deep eutectic solvents and their applications as green solvents, Curr. Opin. Green Sustain. Chem. 21 (2020) 27–33.
[28] R. Stefanovic, G.B. Webber, A.J. Page, Polymer solvation in choline chloride deep eutectic solvents modulated by the hydrogen bond donor, J. Mol. Liq. 279 (2019) 584–593.
[29] N.M. Abbasi, M.Q. Farooq, J.L. Anderson, Investigating the variation in solvation interactions of choline chloride-based deep eutectic solvents formed using different hydrogen bond donors, ACS Sustain. Chem. Eng. 9 (2021) 11970–11980.
[30] P. Xu, Y. Xu, X.F. Li, B.Y. Zhao, M.H. Zong, W.Y. Lou, Reduction of 3-chloropropiophenone with immobilized Acetobacter sp. CCTCC M209061 cells by using deep eutectic solvents as cosolvents, ACS Sustain. Chem. Eng. 3 (2015) 718–724.
[31] F. Chemat, H. Anjum, A.M. Shariff, P. Kumar, T. Murugesan, Thermal and physical properties of (Choline chloride + urea +l-arginine) deep eutectic solvents, J. Mol. Liq. 218 (2016) 301–308.
[32] Q. Chen, L. Chaihu, X. Yao, X. Cao, W. Bi, J. Lin, D.D.Y. Chen, Molecular property-tailored soy protein extraction process using a deep eutectic solvent, ACS Sustain. Chem. Eng. 9 (2021) 10083–10092.
[33] Q. Li, J.Q. Ren, Q. Li, J.H. Di, C. Ma, Y. He, Sustainable conversion of biomass-derived *d*-xylose to furfuryl alcohol in a deep eutectic solvent–water system, ACS Sustain. Chem. Eng. 9 (2021) 10299–10308.
[34] A.K. Schweiger, N. Ríos-Lombardía, C.K. Winkler, S. Schmidt, F. Morís, W. Kroutil, J. González-Sabín, R. Kourist, Using deep eutectic solvents to overcome limited substrate solubility in the enzymatic decarboxylation of bio-based phenolic acids, ACS Sustain. Chem. Eng. 7 (2019) 16364–16370.
[35] R. Fotiadou, M.G. Bellou, K. Spyrou, F. Yan, P. Rudolf, D. Gournis, H. Stamatis, Effect of deep eutectic solvents on the biocatalytic properties of β-glucosidase@ ZnOFe nano-biocatalyst, Sustain. Chem. Pharm. 30 (2022), 100886.
[36] A. Özdemir, E. Şahin, Efficient bioreduction of cyclohexyl phenyl ketone by *Leuconostoc pseudomesenteroides* N13 biocatalyst using a distance-based design-focused optimization model, Mol. Catal. 528 (2022) 112474.
[37] Y. Ni, Z. Mao, R.R. Chen, Outer membrane mutation effects on UDP-glucose permeability and whole-cell catalysis rate, Appl. Microbiol. Biotechnol. 73 (2006) 384–393.
[38] C.C.C.R. De Carvalho, Enzymatic and whole cell catalysis: Finding new strategies for old processes, Biotechnol. Adv. 29 (2011) 75–83.
[39] L. Pan, Q. Li, Y. Tao, C. Ma, H. Chai, Y. Ai, Y.C. He, An efficient chemoenzymatic strategy for valorisation of corncob to furfuryl alcohol in CA:Betaine-water, Ind. Crops Prod. 186 (2022) 115203.
[40] J. Zhang, J. Zhou, G. Xu, Y. Ni, Stereodivergent evolution of KpADH for the asymmetric reduction of diaryl ketones with para-substituents, Mol. Catal. 524 (2022), 112315.
[41] L. He, W. Ye, Y. Xie, Q. Liu, H. Wang, D. Wei, Efficient biocatalytic synthesis of (*R*)-2-chloro-1-(3,4-difluorophenyl)ethanol by the short-chain dehydrogenase PpKR8 from *Paraburkholderia phymatum* STM815, Org. Process Res. Dev. 26 (2022) 278–287.
[42] Y.C. He, D.P. Zhang, Z.C. Tao, Y. Lu, Y. Ding, F. Liu, Z.Z. Zhu, H. Rui, G.W. Zheng, X. Zhang, Improved biosynthesis of ethyl (*S*)-4-chloro-3-hydroxybutanoate by adding l-glutamine plus glycine instead of NAD^+ in β-cyclodextrin–water system, Bioresour. Technol. 182 (2015) 98–102.
[43] R.A. Sheldon, J.M. Woodley, Role of biocatalysis in sustainable chemistry, Chem. Rev. 118 (2018) 801–838.
[44] X. Guo, J.W. Tang, J.T. Yang, G.W. Ni, F.L. Zhang, S.X. Chen, Development of a practical enzymatic process for preparation of (S)-2-chloro-1-(3,4-difluorophenyl) ethanol, Org. Process Res. Dev. 21 (2017) 1595–1601.
[45] S.E. Hooshmand, S. Kumar, I. Bahadur, T. Singh, R.S. Varma, Deep eutectic solvents as reusable catalysts and promoter for the greener syntheses of small molecules: Recent advances, J. Mol. Liq. 371 (2023) 121013.
[46] D.A. Alonso, S.J. Burlingham, R. Chinchilla, G. Guillena, D.J. Ramón, M. Tiecco, Asymmetric organocatalysis in deep eutectic solvents, Eur. J. Org. Chem. 2021 (2021) 4065–4071.
[47] Y.C. He, Z.C. Tao, X. Zhang, Z.X. Yang, J.H. Xu, Highly efficient synthesis of ethyl (*S*)-4-chloro-3-hydroxybutanoate and its derivatives by a robust NADH-dependent reductase from *E. coli* CCZU-K14, Bioresour. Technol. 161 (2014) 461–464.

[48] O. Longeras, A. Gautier, K. Ballerat-Busserolles, J.M. Andanson, Deep eutectic solvent with thermo-switchable hydrophobicity, ACS Sustain. Chem. Eng. 8 (2020) 12516–12520.
[49] L. Weng, M. Toner, Janus-faced role of water in defining nanostructure of choline chloride/glycerol deep eutectic solvent, Phys. Chem. Chem. Phys. 20 (2018) 22455–22462.
[50] Z. Yang, W. Pan, Ionic liquids: Green solvents for nonaqueous biocatalysis, Enzym. Microb. Technol. 37 (2005) 19–28.
[51] W.J. Jiang, J.B. Zhang, Y.T. Zou, H.L. Peng, K. Huang, Manufacturing acidities of hydrogen-bond donors in deep eutectic solvents for effective and reversible NH3 capture, ACS Sustain. Chem. Eng. 8 (2020) 13408–13417.
[52] H. Sun, R. Xin, D. Qu, F. Yao, Mechanism of deep eutectic solvents enhancing catalytic function of cytochrome P450 enzymes in biosynthesis and organic synthesis, J. Biotechnol. 323 (2020) 264–273.
[53] N. Guajardo, P. Domínguez De María, R. Canales, Integrating biocatalysis with viscous deep eutectic solvents in Lab-On-A-Chip microreactors, ChemSusChem 15 (2022), e202102674.
[54] F. Peng, Q.S. Chen, F.Z. Li, X.Y. Ou, M.H. Zong, W.Y. Lou, Using deep eutectic solvents to improve the biocatalytic reduction of 2-hydroxyacetophenone to (R)-1-phenyl-1,2-ethanediol by Kurthia gibsonii SC0312, Mol. Catal. 484 (2020), 110773.
[55] N. Das, T. Khan, N. Subba, P. Sen, Correlating Bromelain's activity with its structure and active-site dynamics and the medium's physical properties in a hydrated deep eutectic solvent, Phys. Chem. Chem. Phys. 23 (2021) 9337–9346.
[56] S. Mao, Evaluation of deep eutectic solvents as co-solvent for steroids 1-en-dehydrogenation biotransformation by *Arthrobacter* simplex, J. Chem. Technol. Biotechnol. 91 (2016) 1099–1104.
[57] Z. Zhu, S. Bi, N. Ye, P. Wang, Production of (*S*)-1-[3,5-bis(trifluoromethyl)phenyl] ethanol via whole-cell catalyst in deep-eutectic solvent-containing micro-aerobic medium system, Molecules 28 (2020) 1855.
[58] J. Cao, R. Wu, Q. Dong, L. Zhao, F. Cao, E. Su, Effective release of intracellular enzymes by permeating the cell membrane with hydrophobic deep eutectic solvents, ChemBioChem 21 (2020) 672–680.
[59] Q. Xia, Y. Liu, J. Meng, W. Cheng, W. Chen, S. Liu, Y. Liu, J. Li, H. Yu, Multiple hydrogen bond coordination in three-constituent deep eutectic solvents enhances lignin fractionation from biomass, Green Chem. 20 (2018) 2711–2721.
[60] Y.C. He, C.X. Jiang, J.W. Jiang, J.H. Di, F. Liu, Y. Ding, Q. Qing, C.L. Ma, One-pot chemo-enzymatic synthesis of furfuralcohol from xylose, Bioresour. Technol. 238 (2017) 698–705.
[61] Q. Li, J. Di, X. Liao, J. Ni, Q. Li, Y.C. He, C. Ma, Exploration of benign deep eutectic solvent–water systems for the highly efficient production of furfurylamine from sugarcane bagasse via chemoenzymatic cascade catalysis, Green Chem. 23 (2021) 8154–8168.
[62] L. Qin, J. Di, Y. He, Efficient synthesis of furfuryl alcohol from corncob in a deep eutectic solvent system, Processes 10 (2022) 1873.
[63] Y. Ni, J. Zhou, Z. Sun, Production of a key chiral intermediate of Betahistine with a newly isolated *Kluyveromyces* sp. in an aqueous two-phase system, Process Biochem. 47 (2012) 1042–1048.
[64] A. Schon, B.R. Clarkson, M. Jaime, E. Freire, Temperature stability of proteins: Analysis of irreversible denaturation using isothermal calorimetry, Proteins 85 (2017) 2009–2016.
[65] T.H. Fatoki, Effect of pH on structural dynamics of HMG-CoA reductase and binding affinity to β-sitosterol, J. Biomol. Struct. Dyn. 41 (2023) 4398–4404.
[66] Q. Yan, X. Zhang, Y. Chen, B. Guo, P. Zhou, B. Chen, Q. Huang, J.B. Wang, From semirational to rational design: Developing a substrate-coupled system of glucose dehydrogenase for asymmetric synthesis, ACS Catal. 12 (2022) 6746–6755.
[67] J.I. Nishigaki, T. Ishida, T. Honma, M. Haruta, Oxidation of β-nicotinamide adenine dinucleotide (NADH) by Au cluster and nanoparticle catalysts aiming for coenzyme regeneration in enzymatic glucose oxidation, ACS Sustain. Chem. Eng. 8 (2020) 10413–10422.
[68] K. Niu, T. Xiong, H.B. Qin, H. Wu, Z.Q. Liu, Y.G. Zheng, 3-Hydroxypropionic acid production by recombinant Escherichia coli ZJU-3HP01 using glycerol–glucose dual-substrate fermentative strategy, Biotechnol. Appl. Biochem. 64 (2017) 572–578.
[69] J. Wang, Z.H. Zhu, M.W. Chen, Q.A. Chen, Y.G. Zhou, Catalytic biomimetic asymmetric reduction of alkenes and imines enabled by chiral and regenerable NAD(P)H models, Angew. Chem. 131 (2019) 1827–1831.
[70] Q. Zhao, Y. Li, B. Shen, Q. Zhao, L. Zhu, L. Jiang, UiO-66-mediated light-driven regeneration of intracellular NADH in Clostridium tyrobutyricum to strengthen butyrate production, ACS Sustain. Chem. Eng. 11 (2023) 3405–3415.
[71] Z. Tang, Q. Li, J. Di, C. Ma, Y.C. He, An efficient chemoenzymatic cascade strategy for transforming biomass into furfurylamine with lobster shell-based chemocatalyst and mutated ω-transaminase biocatalyst in methyl isobutyl ketone-water, Bioresour. Technol. 369 (2023) 128424.
[72] L.J. Wang, C.X. Li, Y. Ni, J. Zhang, X. Liu, J.H. Xu, Highly efficient synthesis of chiral alcohols with a novel NADH-dependent reductase from *Streptomyces coelicolor*,, Bioresour. Technol. 102 (2011) 7023–7028.
[73] C.Y. Weng, X.F. Gao, H.T. Liu, R.L. Chu, W.B. Xie, Y.J. Wang, Y.G. Zheng, Protein engineering of carbonyl reductase for asymmetric synthesis of ticagrelor precursor (1*S*)-2-chloro-1-(3,4-difluorophenyl)ethanol, Biochem. Eng. J. 187 (2022) 108600.
[74] F.J. Zhao, Y. Liu, X.Q. Pei, C. Guo, Z.L. Wu, Single mutations of ketoreductase ChKRED20 enhance the bioreductive production of (1S)-2-chloro-1-(3, 4-difluor-ophenyl) ethanol, Appl. Microbiol. Biotechnol. 101 (2017) 1945–1952.
[75] A. Varenikov, M. Gandelman, Synthesis of chiral alpha-trifluoromethyl alcohols and ethers via enantioselective Hiyama cross-couplings of bisfunctionalized electrophiles, Nat. Commun. 9 (2018) 3566.
[76] H. Yin, P.Q. Luan, Y.F. Cao, J. Ge, W.Y. Lou, Coupling metal and whole-cell catalysis to synthesize chiral alcohols, Bioresour. Bioprocess. 9 (2022) 19.
[77] T. Luo, Z. Dou, Z. Sun, X. Chen, Y. Ni, G. Xu, A novel and robust 3-quinuclidinone reductase from Kaistia algarum for efficient synthesis of (R)-3-quinuclidinol without external cofactor, Mol. Catal. 514 (2021), 111861.